Structural Change in a Multisector Model of Growth

Ngai and Pissarides (2007): Sectoral Productivity, Structural Change, and Balanced Growth

Paper presentation
Group 4
American Economic Review, 2007

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Advanced Macroeconomics II

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PART 1

Question and Contribution

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Central Question

What does the paper explain?

Sectoral TFP grows at different rates. How does this generate long-run shifts in employment shares, and can those shifts coexist with aggregate balanced growth?

Sectoral facts

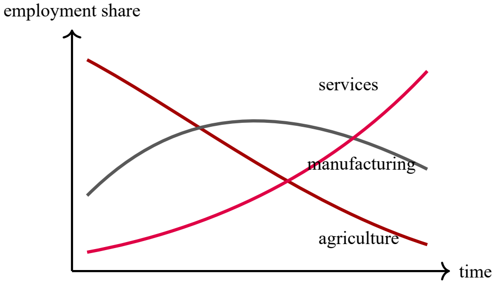
- Employment shares move over long horizons.
- Low-productivity-growth sectors become relatively expensive.
- Nominal shares move more than real consumption shares.

Source: Ngai and Pissarides, American Economic Review, 2007.

Aggregate facts

- Output, consumption, and capital can grow together.
- Aggregate ratios may remain roughly constant.
- The goal is to reconcile reallocation with Kaldor facts.

Structural Change Is Not Aggregate Noise



Evidence discussed in the paper: Kuznets (1966), Maddison (1980), Baumol et al. (1985).

- Kuznets and Maddison document a long-run shift from agriculture toward manufacturing and services.
- Baumol's cost disease predicts rising relative prices in stagnant sectors.
- Ngai and Pissarides emphasize a technological explanation: unequal sectoral TFP growth is enough to move labor shares.

Position in the Literature

Approach	Source of structural change	Relation to this paper
Nonhomothetic preferences	Different income elasticities change demand composition as income grows.	This paper does not rely on Stone-Geary preferences or sector-specific income elasticities.
Technology-based explanation	Different sectoral TFP growth rates change relative prices and expenditure shares.	This is the paper's main mechanism under CES final goods and homothetic preferences.
Balanced-growth restrictions	Preferences and technologies are often tied together by special parameter restrictions.	Ngai and Pissarides keep preference and technology parameters largely separate.

One-sentence contribution

If final goods are poor substitutes, labor moves toward low-TFP-growth sectors; if intertemporal utility is logarithmic, this movement is consistent with aggregate balanced growth.

Presentation Roadmap

Facts sectoral shifts → Model CES goods + TFP gaps → Mechanism prices and labor shares → Aggregate balanced growth → Extensions inputs and capital goods

First: structural change

Derive how employment shares respond to relative prices and sectoral TFP growth rates.

Then: aggregate growth

Identify when structural change is compatible with constant aggregate ratios.

Reading Priorities

1. **Relative prices**: static efficiency gives $p_i/p_m = A_m/A_i$, mapping TFP gaps into price movements.
2. **Substitution elasticity**: the sign of $1 - \varepsilon$ determines the direction of employment reallocation.
3. **Aggregate balance**: structural change makes $\bar{\gamma}$ time-varying; $\theta = 1$ removes this term from the Euler equation.
4. **Robustness**: intermediate inputs and multiple capital goods preserve the logic under Cobb-Douglas aggregation restrictions.

Target takeaway

A sector can absorb more employment precisely because it is technologically stagnant, without forcing aggregate growth to become unbalanced.

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PART 2

Baseline Model

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Environment: Many Consumption Goods, One Capital-Goods Sector

Sectoral structure

- There are m sectors.
- Sectors $i = 1, \ldots, m - 1$ produce consumption goods only.
- Sector m is manufacturing.
- Manufacturing produces both a final good and the capital good.

Key modeling choices

- Labor and capital move freely across sectors.
- Production functions differ only in TFP levels and growth rates.
- Preferences are homothetic CES, not nonhomothetic.

$$U = \int_0^\infty e^{-\rho t} v(c_1, \ldots, c_m)\, dt.$$

Resource Constraints

Factor allocation

$$\sum_{i=1}^{m} n_i = 1, \qquad \sum_{i=1}^{m} n_i k_i = k.$$

Consumption sectors

$$c_i = F^i(n_i k_i, n_i), \quad i \neq m.$$

Manufacturing sector

$$\dot{k} = F^m(n_m k_m, n_m) - c_m - (\delta + \nu)k.$$

Interpretation: non-manufacturing output is consumed immediately; manufacturing output is split between current consumption and investment.

Static Efficiency: TFP Gaps Become Relative Prices

Sectoral technologies

$$F^i = A_i F(n_i k_i, n_i), \qquad \dot{A}_i / A_i = \gamma_i.$$

Static efficiency implies

$$k_i = k, \qquad \frac{p_i}{p_m} = \frac{v_i}{v_m} = \frac{A_m}{A_i}, \quad \forall i.$$

- Capital-labor ratios equalize because factor markets are frictionless.
- A faster-growing sector becomes relatively cheaper over time.
- This is the bridge from technology differences to demand reallocation.

CES Preferences: Two Elasticities

$$v(c_1, \dots, c_m) = \frac{\phi(c_1, \dots, c_m)^{1-\theta} - 1}{1-\theta},$$

$$\phi = \left(\sum_{i=1}^{m} \omega_i c_i^{(\varepsilon-1)/\varepsilon} \right)^{\varepsilon/(\varepsilon-1)}.$$

ε: substitution across goods

Determines whether rising relative prices increase or reduce a sector's expenditure and employment shares.

$1/\theta$: intertemporal substitution

Determines whether structural change can coexist with a balanced aggregate path.

Expenditure Ratio: The Key State for Sectoral Demand

Consumption expenditure of sector i relative to manufacturing

$$\frac{p_i c_i}{p_m c_m} = \left(\frac{\omega_i}{\omega_m}\right)^{\varepsilon} \left(\frac{A_m}{A_i}\right)^{1-\varepsilon} \equiv x_i.$$

If $\varepsilon < 1$

A relative price increase raises expenditure share because demand is price inelastic.

If $\varepsilon > 1$

A relative price increase lowers expenditure share because consumers substitute away strongly.

Aggregate Variables in Manufacturing Units

$$c \equiv \sum_{i=1}^{m} \frac{p_i}{p_m} c_i, \qquad y \equiv \sum_{i=1}^{m} \frac{p_i}{p_m} F^i.$$

Two useful simplifications

$$c = c_m X, \qquad y = A_m F(k, 1), \qquad X \equiv \sum_{i=1}^{m} x_i.$$

- x_i / X is sector i's consumption expenditure share.
- c/y is the economy-wide employment share used for consumption goods.
- $1 - c/y$ is the manufacturing labor share used to produce capital goods.

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PART 3

Structural Change Mechanism

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Employment Shares: The Central Decomposition

Consumption-only sectors

$$n_i = \frac{x_i}{X}\left(\frac{c}{y}\right), \qquad i \neq m.$$

Manufacturing

$$n_m = \frac{x_m}{X}\left(\frac{c}{y}\right) + \left(1 - \frac{c}{y}\right).$$

- $\frac{x_i}{X}$ allocates consumption demand across final goods.
- $\frac{c}{y}$ is the labor share needed for total consumption.
- Manufacturing also supplies investment demand, captured by $1 - c/y$.

Proposition 1: Relative Employment and Relative Prices

Relative employment growth

$$\frac{\dot{n}_i}{n_i} - \frac{\dot{n}_j}{n_j} = (1-\varepsilon)(\gamma_j - \gamma_i), \qquad i,j \neq m.$$

Relative price growth

$$\frac{\dot{p}_i}{p_i} - \frac{\dot{p}_j}{p_j} = \gamma_j - \gamma_i.$$

Direct implication

$$\frac{\dot{n}_i}{n_i} - \frac{\dot{n}_j}{n_j} = (1-\varepsilon)\left(\frac{\dot{p}_i}{p_i} - \frac{\dot{p}_j}{p_j}\right).$$

Derivation sketch

The Sign of $1-\varepsilon$ Determines the Direction

$\varepsilon < 1$

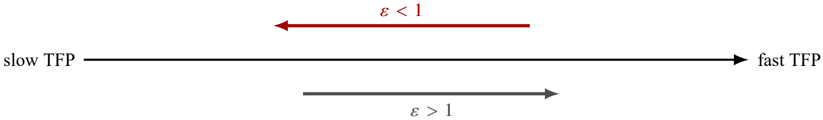
- Goods are poor substitutes.
- Demand is price inelastic.
- Labor moves toward slow-TFP-growth sectors.

$\varepsilon = 1$

- Cobb-Douglas aggregator.
- Expenditure shares are constant.
- TFP gaps change prices but not employment shares.

$\varepsilon > 1$

- Goods are strong substitutes.
- Demand is price elastic.
- Labor moves toward fast-TFP-growth sectors.



Proposition 2: When Do Employment Shares Change?

Case	Condition for structural change	Type of reallocation
Equal sectoral TFP growth	$\dot{c}/c \neq \dot{y}/y$	Between the aggregate consumption sector and the manufacturing capital-goods role.
Different sectoral TFP growth on a balanced path	$\varepsilon \neq 1$ and at least one $\gamma_i \neq \gamma_m$	Across consumption sectors with different TFP growth rates.
Balanced path with $\varepsilon < 1$	Goods are poor substitutes	Employment moves from high-TFP-growth sectors to low-TFP-growth sectors.
Balanced path with $\varepsilon > 1$	Goods are strong substitutes	Employment moves from low-TFP-growth sectors to high-TFP-growth sectors.

Why this matters

Structural change is not a residual. In the model it is pinned down by relative productivity growth and the final-goods substitution elasticity.

Nominal Shares Move More Than Real Shares

Nominal side

- Employment shares equal nominal output shares.
- With $\varepsilon < 1$, low-growth sectors get rising nominal shares.
- This matches the logic of Baumol's cost disease.

Real side

$$\frac{\dot{c}_i}{c_i} - \frac{\dot{c}_j}{c_j} = \varepsilon(\gamma_i - \gamma_j).$$

Small ε dampens changes in real consumption shares.

The paper links this implication to Kravis, Heston, and Summers (1983) and related evidence on services.

Individual Employment Dynamics

Consumption-sector law of motion

$$\frac{\dot{n}_i}{n_i} = \frac{(\dot{c/y})}{c/y} + (1-\varepsilon)(\bar{\gamma} - \gamma_i), \qquad i \neq m,$$

where $\bar{\gamma} = \sum_i (x_i/X)\gamma_i$ is the consumption-share-weighted average TFP growth rate.

Balanced path

If c/y is constant, a sector expands when its TFP growth is on the "right" side of $\bar{\gamma}$.

Moving threshold

$\bar{\gamma}$ itself changes because expenditure weights x_i/X evolve with relative productivity.

Mechanism in One Diagram

Sectoral TFP gap $\gamma_i \neq \gamma_j$ → Relative price changes $\dot{p}_i/p_i - \dot{p}_j/p_j$ → CES demand response controlled by ε → Employment-share dynamics $\dot{n}_i/n_i - \dot{n}_j/n_j$

Main intuition for $\varepsilon < 1$

Productivity growth makes a sector cheaper. But if goods are poor substitutes, demand does not rise enough to absorb all extra output, so labor leaves the fast-growing sector.

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PART 4

Aggregate Balanced Growth

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From Structural Change to Aggregate Growth

Additional production restriction

To study a steady aggregate path, the paper specializes the common production function to Cobb-Douglas:

$$F(n_i k_i, n_i) = k_i^{\alpha} n_i, \qquad \alpha \in (0,1).$$

- Hicks-neutral technology with balanced aggregate growth requires a Cobb-Douglas form.
- The aggregate economy is measured in manufacturing-good units.
- Sectoral TFP growth enters the Euler equation through $\bar{\gamma}$.

Proposition 3: Aggregate Dynamics

Given any initial $k(0)$, equilibrium aggregate dynamics satisfy:

Capital accumulation

$$\frac{\dot{k}}{k} = A_m k^{\alpha-1} - \frac{c}{k} - (\delta + \nu).$$

Euler equation

$$\theta \frac{\dot{c}}{c} = (\theta - 1)(\gamma_m - \bar{\gamma}) + \alpha A_m k^{\alpha-1} - (\delta + \rho + \nu).$$

The new term is $(\theta - 1)(\gamma_m - \bar{\gamma})$: it is zero in a one-sector model and generally time-varying under structural change.

Efficiency Units and the Ramsey Analogy

Define

$$c_e \equiv cA_m^{-1/(1-\alpha)}, \qquad k_e \equiv kA_m^{-1/(1-\alpha)}, \qquad g_m \equiv \frac{\gamma_m}{1-\alpha}.$$

If

$$\psi \equiv (\theta - 1)(\gamma_m - \bar{\gamma})$$

is constant, the aggregate system becomes a Ramsey-style two-dimensional system:

$$\frac{\dot{c}_e}{c_e} = \frac{\alpha k_e^{\alpha-1} + \psi - (\delta + \nu + \rho)}{\theta} - g_m,$$

$$\dot{k}_e = k_e^{\alpha} - c_e - (g_m + \delta + \nu)k_e.$$

Proposition 4: Conditions for Balanced Growth with Structural Change

Necessary and sufficient conditions

An aggregate balanced growth path with structural change exists if and only if

$$\theta = 1, \qquad \varepsilon \neq 1, \qquad \exists i \in \{1, \dots, m-1\} : \gamma_i \neq \gamma_m.$$

Why $\varepsilon \neq 1$?

Without nonunit substitution across goods, relative price changes do not generate employment-share movements on a balanced path.

Intuition

Why $\theta = 1$?

Log utility makes $\psi = 0$, so the aggregate Euler equation no longer inherits the time-varying $\bar{\gamma}$ term.

Economic Intuition for Log Intertemporal Utility

Balanced growth requirement

Aggregate balanced growth requires consumption to remain a constant fraction of aggregate wealth.

- The relevant real return includes a relative-price component: $\gamma_m - \bar{\gamma}$.
- Under structural change, $\bar{\gamma}$ changes because expenditure weights move across sectors.
- If $\theta \neq 1$, consumption reacts to this changing return, so the aggregate path is not exactly balanced.
- If $\theta = 1$, consumption is independent of the interest rate in the required way; the aggregate economy behaves like a one-sector Ramsey model.

Proposition 5: Long-Run Employment Shares

Let sector l be the limiting sector: the smallest TFP-growth sector if $\varepsilon < 1$, or the largest TFP-growth sector if $\varepsilon > 1$.

Asymptotic allocation on the balanced path

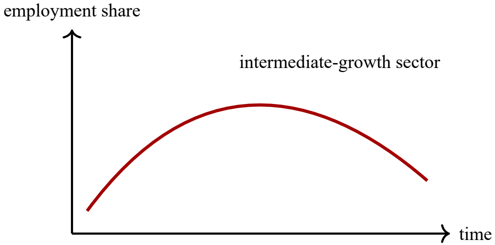
$$n_m^* = \hat{\sigma} = \alpha \left(\frac{\delta + \nu + g_m}{\delta + \nu + \rho + g_m} \right), \qquad n_l^* = 1 - \hat{\sigma}.$$

- Manufacturing retains labor because it produces capital goods.
- The limiting consumption sector absorbs all consumption-goods labor.
- Other consumption sectors may decline monotonically or display hump-shaped employment shares.

Why Hump-Shaped Employment Shares Can Arise

For $\varepsilon < 1$, Lemma A3 gives:

$$\frac{d\bar{\gamma}}{dt} = -(1-\varepsilon)\sum_i \frac{x_i}{X}(\gamma_i - \bar{\gamma})^2 \leq 0.$$



- A sector expands if $\gamma_i < \bar{\gamma}$.
- As $\bar{\gamma}$ declines, fewer sectors satisfy this condition.
- A sector can first expand and then contract once $\bar{\gamma}$ falls below its own TFP growth rate.

What if $\theta \neq 1$?

Exact balanced growth fails

When $\theta \neq 1$, the Euler term

$$\psi = (\theta - 1)(\gamma_m - \bar{\gamma})$$

is time-varying during structural change.

But the deviation is small and asymptotic

$$\dot{\psi} = (\theta - 1)(1 - \varepsilon) \sum_i \frac{x_i}{X} (\gamma_i - \bar{\gamma})^2.$$

Since TFP growth rates are small numbers, this term is second order relative to the employment-share movements.

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PART 5

Extensions and Discussion

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Extension 1: Intermediate Goods

Many sectors sell output to firms as well as households. The extension lets goods be used as intermediate inputs.

Modified production function

$$F^i = A_i n_i k_i^{\alpha} q_i^{\beta}, \qquad \alpha, \beta > 0, \quad \alpha + \beta < 1.$$

Balanced-growth restriction

The intermediate-input aggregator must be Cobb-Douglas:

$$\eta = 1.$$

The paper notes that U.S. input-output data show large intermediate-use shares across sectors.

Intermediate Goods: Employment Shares

Consumption-sector employment

$$n_i = \frac{x_i}{X}\left(\frac{c}{y}\right) + \varphi_i\beta, \qquad i \neq m.$$

Manufacturing employment

$$n_m = \left[\frac{x_m}{X}\left(\frac{c}{y}\right) + \varphi_m\beta\right] + \left(1 - \beta - \frac{c}{y}\right).$$

- The structural-change mechanism still applies to the consumption component.
- The intermediate-input term $\varphi_i\beta$ is a constant employment floor.
- In the limit, intermediate-good producers can retain employment even if their final-consumption component vanishes.

Extension 2: Many Capital Goods

Instead of a single manufacturing capital good, suppose there are κ capital-producing sectors aggregated into G.

Relative employment across capital-goods sectors

$$\frac{n_{m_j}}{n_{m_i}} = \left(\frac{\xi_{m_j}}{\xi_{m_i}}\right)^{\mu} \left(\frac{A_{m_i}}{A_{m_j}}\right)^{1-\mu}.$$

$$\frac{\dot{n}_{m_j}}{n_{m_j}} - \frac{\dot{n}_{m_i}}{n_{m_i}} = (1-\mu)(\gamma_{m_i} - \gamma_{m_j}).$$

The logic mirrors the baseline model, but the relevant substitution elasticity is now μ inside the capital-goods aggregator.

Many Capital Goods: Balanced-Growth Implication

Key result

Multiple capital-producing sectors with different TFP growth rates can coexist on an aggregate balanced growth path only if

$$\mu = 1.$$

- The aggregate capital-sector growth rate is a weighted average of capital-sector TFP growth rates.
- If $\mu \neq 1$, the weights depend on relative productivity levels and therefore change over time.
- If $\mu = 1$, the Cobb-Douglas aggregator fixes expenditure/input shares, so the aggregate growth rate is constant.

Empirical Predictions

Sector type	Prediction when $\varepsilon < 1$
Final consumption sectors	Employment-share growth is negatively related to sectoral TFP growth and positively related to relative price inflation.
Intermediate-goods intensive sectors	The same force operates, but the constant intermediate-input term weakens the clean linear relation.
Capital-goods intensive sectors	If capital-goods aggregation is Cobb-Douglas, employment shares need not have the same relation with relative TFP growth.

Useful empirical distinction

The model predicts different labor-share dynamics depending on whether a sector mainly supplies final consumption, intermediate inputs, or capital goods.

Strengths of the Paper

Theoretical strengths

- Clean separation between preferences and technologies.
- Structural change generated without nonhomothetic preferences.
- Aggregate dynamics reduce to a Ramsey benchmark under $\theta = 1$.

Interpretive strengths

- Gives a precise version of Baumol's mechanism.
- Explains why relative prices and employment can rise together.
- Reconciles large nominal shifts with small real-share shifts.

Limitations and Discussion Questions

- **No nonhomotheticity**: the model intentionally abstracts from income-elasticity channels that are important in other structural-change models.
- **Identical production shapes**: sectoral differences enter mainly through TFP growth, not through capital intensity or fixed factors.
- **Balanced-growth knife edge**: exact coexistence requires $\theta = 1$, though the paper argues deviations may be quantitatively small.
- **Sector classification**: real industries often produce final, intermediate, and capital goods at the same time.

Discussion prompt

Is the low final-goods substitution elasticity more plausible at broad sectoral levels than at detailed industry levels?

Main Takeaways

1. Unequal sectoral TFP growth changes relative prices.
2. CES demand converts relative price changes into expenditure-share changes.
3. If $\varepsilon < 1$, labor moves toward sectors with lower TFP growth.
4. Aggregate balanced growth can coexist with this movement if $\theta = 1$.
5. Intermediate inputs and multiple capital goods preserve the core idea under Cobb-Douglas aggregation restrictions.

Bottom line

Structural change can be a permanent feature of sectoral allocation while the aggregate economy still looks balanced.

Q&A

Thank You

Questions and comments are welcome

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PART 6

Appendix

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Derivation Sketch: Relative Employment

Start from

$$n_i = \frac{x_i}{X}\frac{c}{y}, \qquad x_i = \left(\frac{\omega_i}{\omega_m}\right)^{\varepsilon}\left(\frac{A_m}{A_i}\right)^{1-\varepsilon}.$$

For two consumption sectors $i, j \neq m$:

$$\begin{aligned}\frac{\dot{n}_i}{n_i} - \frac{\dot{n}_j}{n_j} &= \frac{\dot{x}_i}{x_i} - \frac{\dot{x}_j}{x_j} \\ &= (1-\varepsilon)(\gamma_m - \gamma_i) - (1-\varepsilon)(\gamma_m - \gamma_j) \\ &= (1-\varepsilon)(\gamma_j - \gamma_i).\end{aligned}$$

Back

Why $\theta = 1$ Removes the Aggregate Distortion

The aggregate Euler equation is

$$\theta \frac{\dot{c}}{c} = (\theta - 1)(\gamma_m - \bar{\gamma}) + \alpha A_m k^{\alpha - 1} - (\delta + \rho + \nu).$$

During structural change, expenditure weights x_i / X move, so

$$\bar{\gamma} = \sum_i \frac{x_i}{X} \gamma_i$$

generally changes over time.

Implication

Exact balanced growth requires the time-varying term to disappear. Setting $\theta = 1$ makes $(\theta - 1)(\gamma_m - \bar{\gamma}) = 0$.

Back